

SoC and memory interleaving: (part 1)

সিদ্ধান্ত mainly processor এর নিয়ন্ত্রণ কোন কোন RAM connect করলে processor বি-ওর ~~বাক~~ run করতে পারে।

CPU এর সাথে Direct connect করে RAM & Ram

$$\text{memory capacity} = 2^{AB} \cdot DB \text{ bit}$$

$$= \frac{2^{AB} \cdot DB}{8} \text{ Byte}$$

$$= \frac{2^{AB} \cdot DB}{2^{20} \cdot 8} \text{ MB}$$

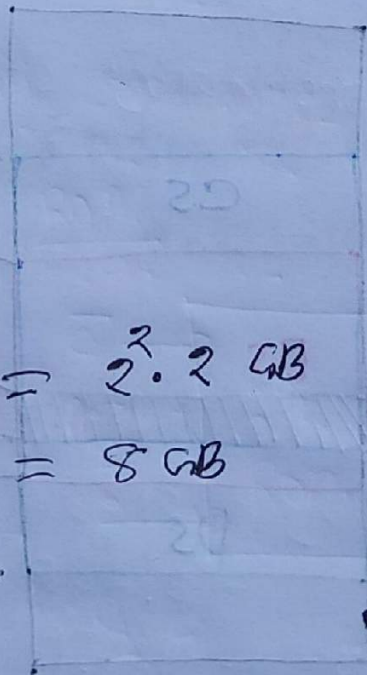
slide pg : 14

Address Bus = 32 bit

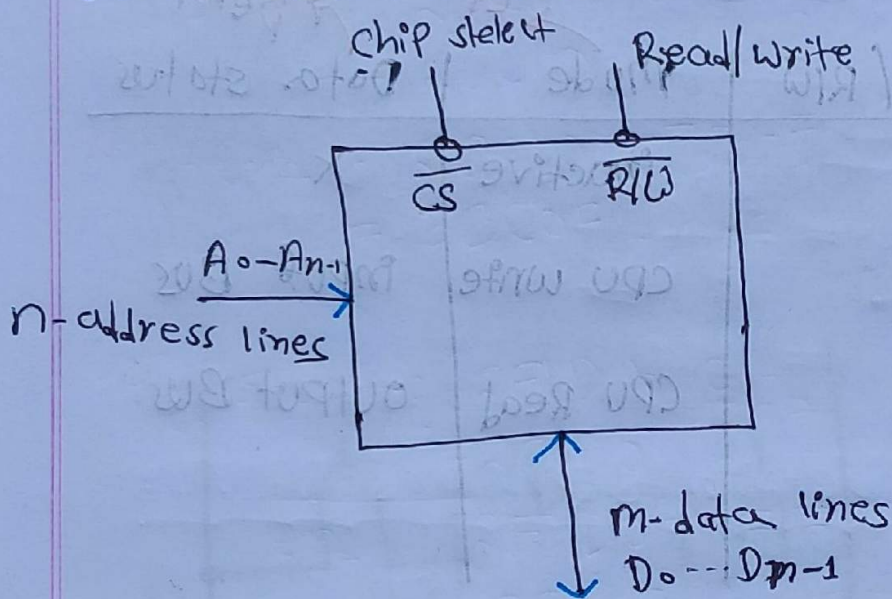
Data Bus = 16 bit

$$\text{capacity} = \frac{2^{32} \cdot 16}{2^{30} \cdot 8} = 2^2 \cdot 2 \text{ GB}$$

$$= 8 \text{ GB}$$



RAM / memory organization:



~~CS~~ →

CS → chip select

CE → chip enable

$\left. \begin{array}{l} 1; \text{ chip selected / enable} \\ 0; \text{ chip disable / dead} \end{array} \right\}$

N.B At a time atmost 1 RAM chip can be enable

$\overline{WE} / R/\overline{W}$

→ 1; Read

→ CPU read and Data Bus outgoing (RAM → CPU)

→ 0; Write

→ CPU Write and Data Bus incoming (CPU → RAM)

Operation of RAM chip

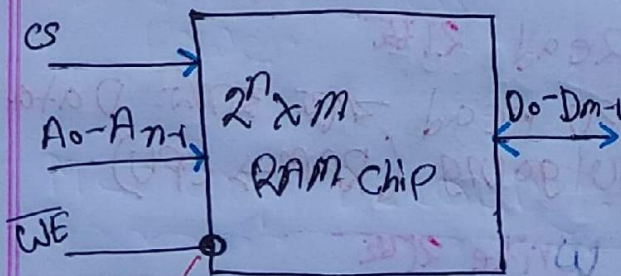
CS	we/R $\bar{w}$	mode	Data status
0	X	inactive	X
1	0	cpu write	input Bus
1	1	cpu Read	output Bus

CS = 0, আমি সব সেকা 50 ফিল্ডে লক্ষ্য রাখা না

CS = 1, আমি ~~সব~~ সব ডাটাবই Action কি হবে তা
খালি রাখা-আমি খালি রাখা (মধ্যে 9 হবে।

Typical RAM chip:

RAM



আমি আমি সব
input মাঝে

ROM

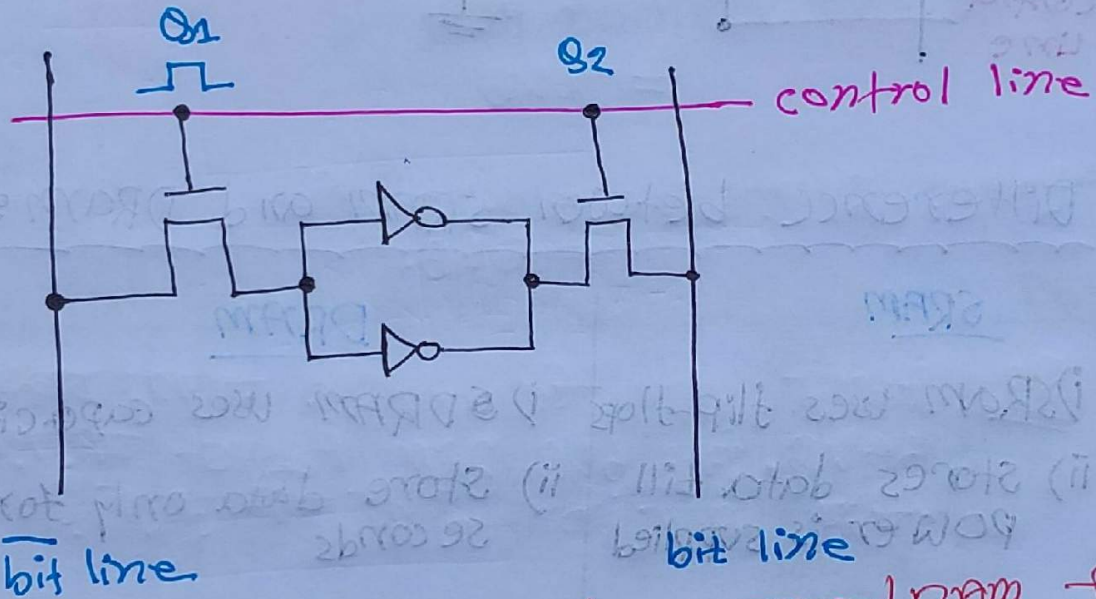


ROM চ কিছু Write করা
মাঝে না জাই আমি Read

Two types of RAMs:

i) Static RAM (SRAM)

SRAM faster than DRAM



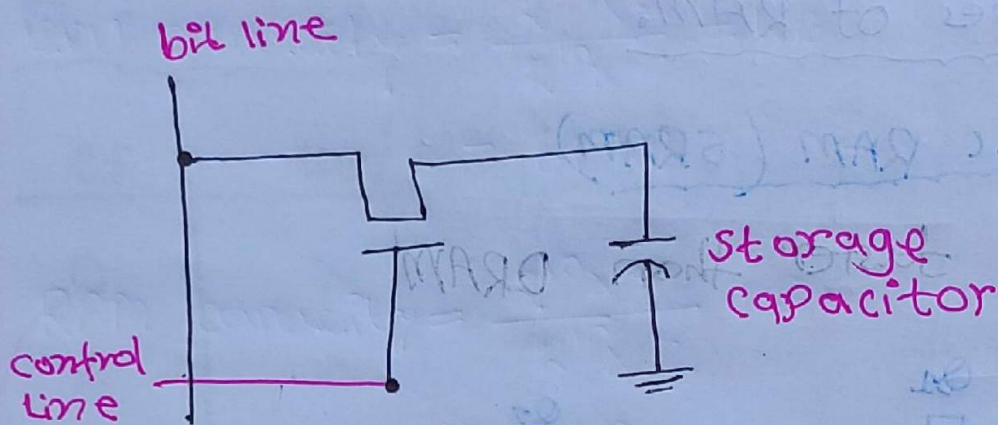
Structure for mainly memory / RAM for 1 bit Data & address - use 24.

ii) Dynamic RAM (DRAM)

Consist mainly of capacitor and transistor.

Capacitor for mainly storage of data and transistor. And transistor capacitor charge for certain time and then refresh. Second write charge refresh. Write refresh. DRAM slow than SRAM.

Write refresh



Difference between SRAM and DRAM:

SRAM

- i) SRAM uses flip-flops
- ii) stores data till power is supplied
- iii) Does not refresh the memory cell
- iv) consume less power

DRAM

- i) DRAM uses capacitors
- ii) store data only for few seconds
- iii) continuous refreshing is required for retain data
- iv) consume more power

Memory Decoding:

Basically, two techniques are used to decode the address

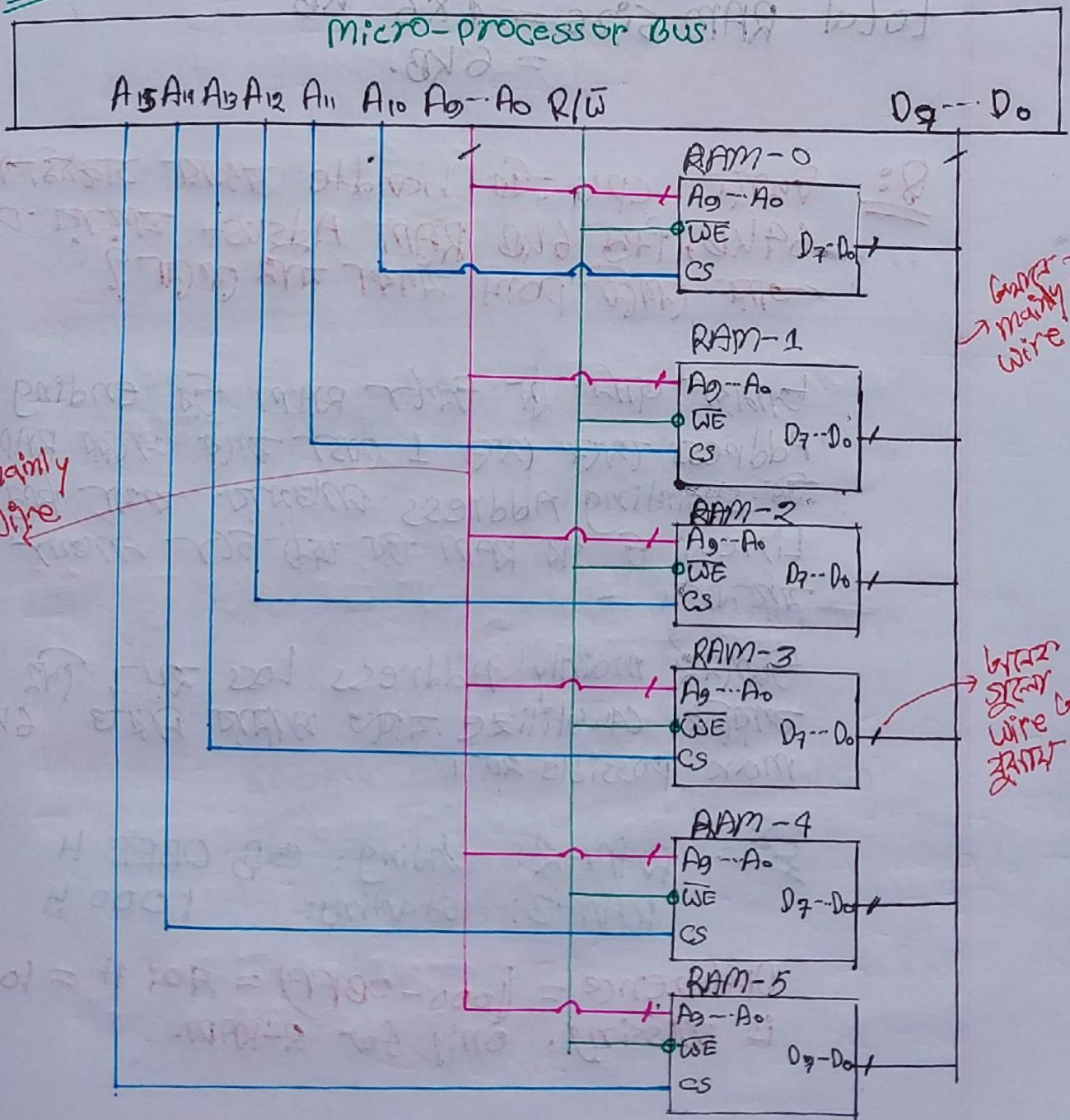
- i) Linear or partial decoding
- ii) Absolute or Full

i) Linear/partial decoding :

processor ke 16^{th} Bit AB

→ 10 ke RAM ke 10 ke AB ke (A₀-A₉)
 → 6 ke RAM connect karta hai (A₁₀-A₁₅)

Ex:



mainly 10 bit wire connect

mainly 6 bit wire

mainly 6 bit wire connect

$$\text{CPU capacity} = \frac{2^{16} \times 8}{2^{10} \times 8} = \text{KB}$$

$$= 64 \text{ KB}$$

$$\text{RAM capacity} = \frac{2^{10} \times 8}{2^{10} \times 8} \text{ KB}$$

$$= 1 \text{ KB}$$

ଯଦି କଲିବ 7KB
RAM Use
ସମ୍ଭବ ନୁହେଁ
ସୁଲଭ 6KB
ସି ଯାହା possible
ନା।

$$\text{total RAM size} = 1 \times 6 \text{ KB}$$

$$= 6 \text{ KB.}$$

Q: ଚିହ୍ନଟ କର CPU ର handle ସହଜରେ
64KB, ମିଳୁ 6KB RAM Adjust ସହଜରେ କର
ଆଉ କୌଣସି port ସଂଖ୍ୟା ନାହିଁ କିମ୍ବା?

ଆମର କାମ ହେଉଛି RAM ର ending
Address ଖୋଜିବା ଯାହା 1 କରୁଥିବା ସମୟରେ RAM
ର starting Address ଖୋଜିବା ସହଜ, କିନ୍ତୁ
Linear ଓ ଅନ୍ୟ RAM ରି ଫର ଚଳେ ନାହିଁ
ସହଜ।

ଚିହ୍ନଟ ମାଧ୍ୟମରେ Address Loss ହୁଏ, ମିଳୁ
ସହଜରେ utilize ସହଜ ମାଧ୍ୟମରେ କିନ୍ତୁ 6KB
max possible ହୁଏ।

Ex: RAM-2: ending: 0BFF H
RAM-3: starting: 1000 H

Difference = (1000 - 0BFF) = 401 H = 1025 index
is missing. only for 2-RAM.

Memory mapped/Range:

RAM-1:

start: $\overline{000001} \overline{000000} \overline{000000}$
= 0900 H

end: $\overline{000001} \overline{111111} \overline{111111}$
= 07FF H

OR
end = 0900 + 3FF
= 07FF H

Starting memory cs connected to pin 1
আবশ্যকীয় স্ট্রাকচার 200

ending of cs pin size
উপস্থিত সব 1 হবে।

RAM-2:

start: $\overline{000010} \overline{000000} \overline{000000}$
= 0800 H

end: = 0800 + 3FF
= 0BFF H

RAM-3:

start: $\overline{000100} \overline{000000} \overline{000000}$
= 1000 H

end = 1000 + 03FF
= 13FF H

RAM-4:

start = $\overline{001000} \overline{000000} \overline{000000}$
= 2000 H

end = 2000 + 03FF
= 23FF H

RAM-6:

start = $\overline{010000} \overline{000000} \overline{000000}$
= 8000 H

end = 8000 + 03FF
= 83FF H

Fall 25 (94(a))

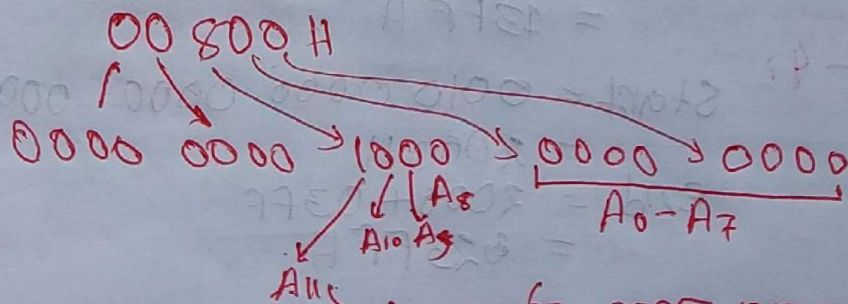
Microprocessor $\rightarrow$ AB = 20 bit
DB = 32 bit
RAM = 5 KB

RAM chip $\rightarrow$ AB = 8 bit
DB = 32 bit

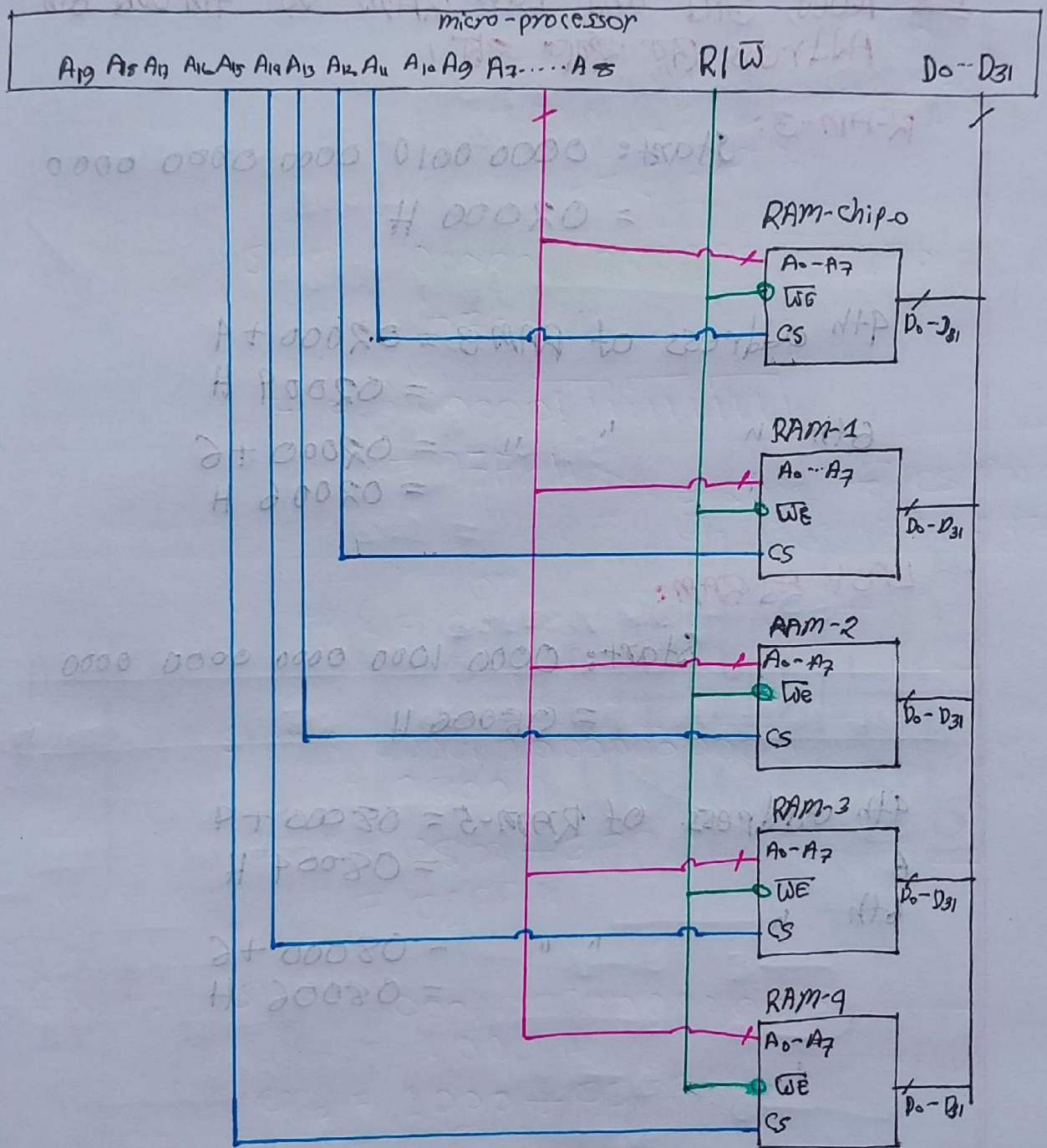
$$\text{size} = \frac{2^8 \cdot 32}{8 \times 2^{10}} \text{ KB} \\ = 1 \text{ KB}$$

$$\text{so total RAM chip needed} = \frac{5}{1} \text{ KB} \\ = 5 \text{ RAM}$$

N.B Mainly RAM সুলো বই (খোঁজ) করে
চলান আছে।



অর্থাৎ RAM
connect এর AH ও



N.B Now, 3rd and Last RAM for 4th and 6th Address (যে বসে হবে)।

RAM-3:

$$\text{Start: } 0000\ 0010\ 0000\ 0000\ 0000 \\ = 02000\text{ H}$$

$$\begin{aligned} \text{4th address of RAM-3} &= 02000 + 4 - 1 \\ &= 02003\text{ H} \end{aligned}$$

$$\begin{aligned} \text{6th address of RAM-3} &= 02000 + 6 - 1 \\ &= 02005\text{ H} \end{aligned}$$

LAST RAM:

$$\begin{aligned} \text{Start: } 0000\ 1000\ 0000\ 0000\ 0000 \\ = 08000\text{ H} \end{aligned}$$

$$\begin{aligned} \text{4th address of RAM-5} &= 08000 + 4 - 1 \\ &= 08003\text{ H} \end{aligned}$$

$$\begin{aligned} \text{6th address of RAM-5} &= 08000 + 6 - 1 \\ &= 08005\text{ H} \end{aligned}$$

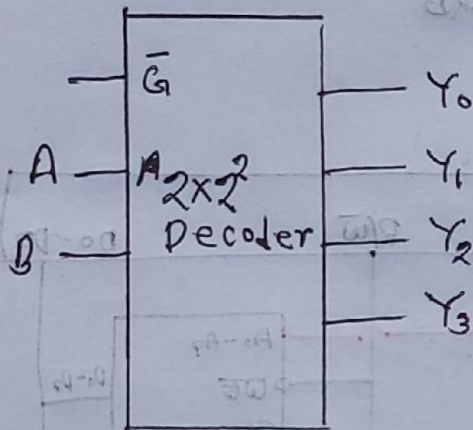
ii) Absolute or full decoding:

Linear decoding problem solve ~~কার~~ full decoding use ~~কার~~ ~~সহ~~।

এর মূল্য ~~সহ~~ 2×2^2 decoder use ~~কার~~ ~~সহ~~।

Decoder:

Truth Table of decoder



G	A	B	Y_0	Y_1	Y_2	Y_3
1	X	X	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	0

N.B

$Y_0 \dots Y_3 \rightarrow 1$ ~~কার~~ OFF, 0 ~~কার~~ ON

$G=1 \rightarrow$ ~~কার~~ input, output ~~কার~~, ~~কার~~ circuit shutdown

এর $Y_0 \dots Y_3 \rightarrow 0$ ~~কার~~ ON, ~~কার~~ RAM CS
1 ~~কার~~ ON, so $Y_0 \dots Y_3$ ~~কার~~ not ~~কার~~ ~~কার~~ CS ~~কার~~ connect ~~কার~~।

$G \rightarrow$ ~~কার~~ ~~কার~~ not ~~কার~~, output not ~~কার~~ ~~কার~~ ~~কার~~।

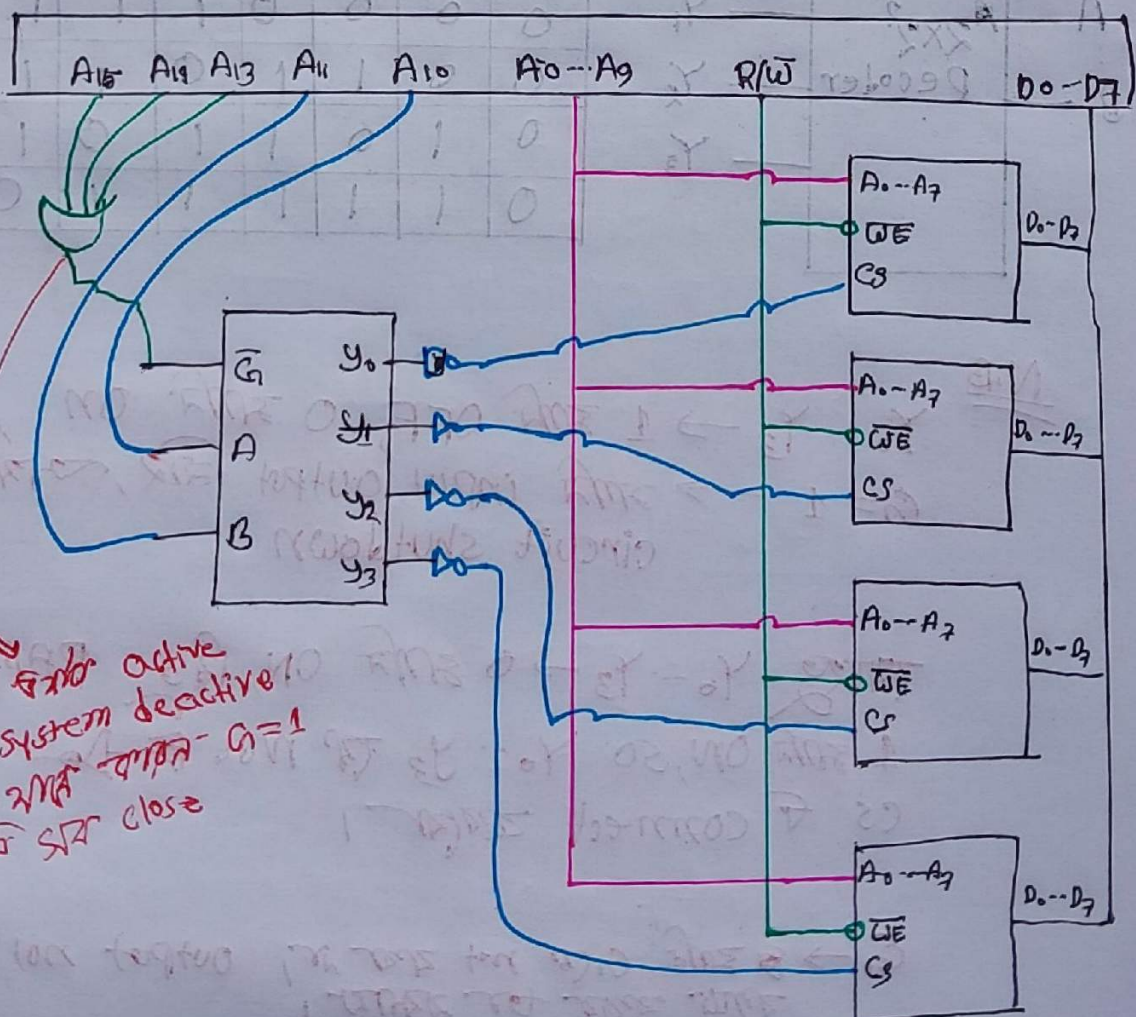
Ex: slide-40

Microprocessor $\rightarrow$ AB = 16 bit
DB = 8 bit
RAM = 4KB

RAM chip $\rightarrow$ AB = 10 bit
DB = 8 bit

$$\text{size} = \frac{2^{10} \times 8}{8 \times 2^{10}} = 1\text{KB}$$

SO, 4 chip needed



Microprocessor active
RAM system deactive
RAM chip - G=1
RAM chip close